

An Introduction to E-Commerce

with a Focus on Machine Learning, Deep Learning, and Artificial Intelligence

Lecture 02: Platforms, Marketplaces, and Business Models

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Session plan (90 minutes)

- ➊ From storefront to platform
- ➋ Value creation, value capture, and growth
- ➌ Governance and trust as system requirements
- ➍ Business model patterns (and architectural implications)
- ➎ Where AI fits (system-level view)
- ➏ Hands-on (placeholder)

By the end of this lecture, you should be able to:

- Distinguish e-commerce *channels* (storefronts) from *platforms* (ecosystems) and *marketplaces* (multi-seller).
- Analyze platform economics: network effects, multi-homing, and pricing of participation.
- Translate business model choices into architectural requirements (identity, trust, data, integration, ERP).

From storefront to platform

- **Storefront:** a single merchant selling to buyers (B2C/B2B) via a web/app channel.
- **Platform:** a system that enables interactions among multiple participant groups (e.g., buyers, sellers, advertisers, logistics providers).
- **Marketplace:** a platform where multiple sellers list items and transactions are mediated by platform policies and infrastructure.

Value creation, value capture, and growth

- **Value creation and matching:** Platforms create value by reducing search and transaction costs, increasing variety, and improving trust; AI frequently strengthens these effects by improving matching and decision support.
- **Value capture models:** Take rate (commission on completed transactions); Subscription (seller or buyer membership tiers)
- **Network effects:** **Direct network effects:** more users attract more users (e.g., social commerce communities).; **Indirect network effects:** more buyers attract more sellers and vice versa (classic marketplace dynamic).

Governance and trust as system requirements

- Seller onboarding, verification, and compliance workflows (KYC/KYB where relevant).
- Catalog integrity: listing policies, moderation, and counterfeit detection.
- Reviews/ratings, dispute resolution, refunds/returns policy enforcement.
- Trust & safety operations: rule+ML detection, case management, and human-in-the-loop escalation.

Business model patterns (and architectural implications)

- **B2C retail:** Key systems: PIM/catalog, promotions, payments, fulfillment, customer support.; ERP/CRM integration: order-to-cash, inventory, accounting, returns.
- **B2B commerce:** Quotes, negotiated pricing, approval workflows, invoicing, credit limits.; Integration-heavy: procurement, supplier management, and EDI/API flows.
- **Marketplace:** Split catalog ownership (platform vs sellers), seller tools, payout/settlement.; Additional "control plane": seller policies, listing moderation, abuse prevention.
- **Digital services (subscriptions):** Recurring billing, entitlements, and churn/retention analytics.; Usage-based pricing requires event design and metering.

Where AI fits (system-level view)

- **Matching layer:** search, recommendations, ranking.
- **Trust layer:** fraud/abuse detection, content moderation.
- **Operations layer:** forecasting, replenishment, routing.
- **Experience layer:** personalization and conversational commerce.

- Create a one-page *platform blueprint*: participant groups, value exchange, pricing model, and required capabilities.
- Map capabilities to an EA view and identify which capabilities are ERP-owned vs commerce-owned.